

Zach Wetmore

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EDUCATION

University of Minnesota, College of Science and Engineering
Bachelor of Science, Mathematics - Actuarial Science Emphasis
Bachelor of Science, Computer Science
Minor, Statistics

Minneapolis, MN

May 2025

GPA: 3.8

TECHNICAL SKILLS & ACTUARIAL EXAMS

Programming & Software Tools: Python, C/C++, C#, Java, JavaScript/TypeScript, OCaml, Matlab, Excel, Docker, SQL
Frameworks: ASP.NET (C#), Angular, Electron, Django, Flask, WinForms
Machine Learning & AI Tools: NumPy, Pandas, scikit-learn, TensorFlow, Generative AI (OpenAI, Cursor, etc)
Version Control & Development Lifecycle: Git/Github, Subversion, Agile Scrum
Actuarial Exams Passed: Financial Math (FM), Probability (P)

WORK EXPERIENCE

Allianz Life

Golden Valley, Minnesota

Actuarial Analyst - Modeling & Systems

May 2024 - Present

- Utilized actuarial math to develop and optimize models within the software development lifecycle in a Scrum environment, focusing on extensibility, maintainability, and efficiency. (RiskAgilityFM/C++, Excel)
- Architected and delivered a centralized Windows Forms application setting the standard for team-wide software, employing OOP design, advanced concurrency control (mutexes, file locks), and optimized data management (lazy/eager loading) to create robust, reusable tools used across the organization.
- Refactored and corrected crediting logic across all actuarial models, including the dedicated CARVM model, improving accuracy, consistency, and reliability for regulatory and internal reporting.

Allianz Life

Golden Valley, Minnesota

Software Engineer - Modeling & Systems

October 2025 - Present

- Developed and maintained full-stack web applications using C# ASP.NET for the backend and Angular for the frontend, delivering robust, user-friendly interfaces..

Colder Products Company (CPC)

Arden Hills, Minnesota

Software Engineering Co-Op

Jan 2024 - April 2024

- Designed and implemented RFID device testing GUIs in C# using Windows Forms, applying a layered architecture similar to MVC to manage XML serialization.

PROJECT EXPERIENCE

Football Scouting Report

- Developed and maintained a Python-based data analysis pipeline, automating ETL processes for complex sports data while ensuring data integrity through robust cleaning and preprocessing routines.
- Applied advanced statistical analysis and visualization techniques to derive actionable insights, leveraging hypothesis testing, and data visualization to enhance decision-making.

Spotify-Apple Music Program

- Developed a multi-threaded Python application to transfer playlists between Apple Music and Spotify. Integrated AppleScript and Spotify's Web API to automate playlist migration and song addition, with additional optimizations for managing API rate limits effectively.

Custom Shell Program, 4041 Introduction to Operating Systems

- Developed a custom Unix shell in C, implementing features such as signal handling, job control, and command execution, enhancing system interaction and process management capabilities.
- Leveraged Docker to containerize the shell application, ensuring consistent execution environments and simplifying deployment across different systems.

LEADERSHIP EXPERIENCE

Big Brothers Big Sisters, Big Brother, 2018 – 2025

Weight Room Instructor, Medford High School, Summer 2020

